

Ramaravind Kommiya Mothilal

PhD Candidate, University of Toronto
Data Sciences Institute (DSI) Graduate Fellow

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Research Objective

My research focuses on AI safety in practice, addressing the computational questions technical practitioners face when working with data and model workflows under the requirements of AI safety. I build methodological frameworks, computational methods, metrics, and datasets, drawing on user studies and the philosophical study of reasoning and argumentation.

Education

Present Sep 2022	University of Toronto PhD in Information ➤ Advisors: Prof. Syed Ishtiaque Ahmed and Prof. Shion Guha ➤ Topic: Drawing on in-depth user studies of how practitioners approach AI responsibly in practice, I focus on nuanced but practically consequential problems at three stages of a practitioner's safety workflow: evaluating the models at the center of their system, evaluating the auxiliary models used to automate judgments, and mitigating model failures under ambiguity about safety.	Toronto, Canada
Feb 2018 Sep 2016	Katholieke Universiteit Leuven (KU Leuven) MSc in Artificial Intelligence (<i>Engineering and Computer Science Specialization</i>) ➤ Percentage: 76.81% (Magna Cum Laude) ➤ Thesis: Statistical Modeling of Students' Performance in an Open-Admission Bachelor Program in Flanders - The need to discriminate between explanatory and predictive modeling [📄]	Leuven, Belgium
May 2015 July 2011	National Institute of Technology (NIT) Trichy Bachelor in Instrumentation and Control Engineering ➤ CGPA: 8.1/10 (First Class)	Tiruchirappalli, India

Experience

Present Aug 2025	University of Toronto Family Housing Residence Advisor ➤ Conducted a hands-on beginner-friendly series to help students, professionals, and lifelong learners build key skills in online writing with LaTeX/Overleaf, managing literary sources, coding with Python, and using AI tools like ChatGPT responsibly [📄]	Toronto, Canada
Jun 2022 Mar 2021	Everwell Health Solutions Data Scientist Mentor: Dr. Bill Thies ➤ Designed a systematic data hygiene framework to anticipate and remove sources of data bias in downstream ML tasks. ➤ Developed indicators to evaluate and improve engagement of health providers to facilitate patient adherence to medications. ➤ Built various dashboards with Power BI and Apache Superset for different stakeholders, including USAID and Central TB Division—Government of India.	Bangalore, India
Feb 2021 Jan 2019	Microsoft Research Center for Societal Impact through Cloud & AI [📄] AI Center Fellow Advisors: Dr. Amit Sharma, Dr. Joyjeet Pal ➤ Developed a general optimization framework—DiCE(Diverse Counterfactual Explanations)—to generate and evaluate a diverse set of hypothetical examples that show people how to obtain a different prediction. [FAccT'20] [📄] [Part of Microsoft's Responsible AI Toolbox 📄] ➤ Studied the complementarities of Feature Attribution and Counterfactual explanation methods to demonstrate the value of considering multiple kinds of model explanations. [AIES'21] ➤ Studied how occupying (un)favourable positions in Indian politicians' social networks leads to inequality in terms of both information received and propagation of new information [CSCW'21]	Bangalore, India
Dec 2018 July 2018	Microsoft Research Technology for Emerging Markets [📄] Research Intern Advisors: Dr. Amit Sharma	Bangalore, India
Jul 2016 Jul 2015	Royal Bank of Scotland (RBS) Data Archival Team Analyst	Chennai, India

Computing Skills

- › **Operating System:** Windows, Linux
- › **Database/Server:** MySQL, PostgreSQL, SQL Server
- › **Language:** Python, R, C/C++, Java, HTML, CSS, JavaScript, SQL
- › **Machine Learning Frameworks:** PyTorch, TensorFlow, JAX
- › **Visualization Software/Tools:** Power BI, Apache Superset, Tableau

Software and Open-Source Contribution

- › **AmbiGuard: Reflective Evaluation of AI Safety Guardrails** : University of Toronto - [sandbox](#)
- › **RaR: A Worksheet for Reflective Use of LLM Reasoning in HCI** : University of Toronto - [framework](#)
- › **ArC: Argument-Based Consistency in Toxicity Explanations of LLMs** : University of Toronto - [commits](#), [codebase](#)
- › **AsAP: Articulating Assumptions in Machine Learning** : University of Toronto - [framework](#)
- › **Non-Ideal Methodological Framework for Responsible ML** : University of Toronto - [framework](#), [visualizer](#)
- › **DiCE: Diverse Counterfactual Explanations for Machine Learning** : Microsoft Research India - [commits](#), [codebase](#)

Awards

- › **The Centre for Ethics (C4E) Graduate Fellowship** - C\$2,500 - 2026
- › **Data Sciences Institute (DSI) Graduate Fellowship** - C\$75,000 - 2024-26
- › **Human-Computer Interaction Consortium (HCIC) - University of Toronto Nominee** - \$650 - 2026
- › **ACM FAccT Conference (Fairness, Accountability, and Transparency) - Travel Grant** - \$750 - 2026
- › **Schwartz Reisman Institute for Technology and Society (SRI) Graduate Fellowship** - C\$7,500 - 2023-24

Publications

P=Preprint, I=In Press, C=Conference, W=Workshop, D=Demo/Poster, J=Journal

- [I.1] **Evaluating Second-Order Bias of LLMs Through Epistemic Entitlement** [🔗]
[Ramaravind Kommiya Mothilal](#), Terry Zhang, Raiyan Ahmed, Zhijing Jin, Shion Guha, Syed Ishtiaque Ahmed
In Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP) [EMNLP 2026]
- [P.1] **Reasoning About Reasoning: Towards Informed and Reflective Use of LLM Reasoning in HCI** [🔗]
[Ramaravind Kommiya Mothilal](#), Sally Zhang, Syed Ishtiaque Ahmed, and Shion Guha
Under Review
- [C.13] **Argument-Based Consistency in Toxicity Explanations of LLMs** [🔗]
[Ramaravind Kommiya Mothilal](#), Joanna Roy, Syed Ishtiaque Ahmed, and Shion Guha
In Proceedings of the 19th Conference of the European Chapter of the ACL (EACL 2026). [EACL 2026]
- [C.12] **Talking About the Assumption in the Room** [🔗]
[Ramaravind Kommiya Mothilal](#), Faisal M Lalani, Syed Ishtiaque Ahmed, Shion Guha, and Sharifa Sultana
2025 CHI Conference on Human Factors in Computing Systems [CHI'25]
- [C.11] **Towards a Non-Ideal Methodological Framework for Responsible ML** [🔗]
[Ramaravind Kommiya Mothilal](#), Shion Guha, Syed Ishtiaque Ahmed
2024 CHI Conference on Human Factors in Computing Systems [CHI'24]
- [C.10] **Caste Capital on Twitter: A Formal Network Analysis of Caste Relations among Indian Politicians** [🔗]
[Ramaravind Kommiya Mothilal](#)*, Palashi Vaghela*, Joyojeet Pal (* = Equal Contribution)
Proceedings of the ACM on Human-Computer Interaction (CSCW) [CSCW'22]
- [C.9] **Voting with the stars: Analyzing Partisan Engagement between Celebrities and Politicians in India** [🔗]
[Ramaravind Kommiya Mothilal](#), Dibyendu Mishra, Sachita Nishal, Faisal M. Lalani, Joyojeet Pal
Proceedings of the ACM on Human-Computer Interaction (CSCW) [CSCW'22]
- [C.8] **Towards unifying feature attribution and counterfactual explanations: Different means to the same end** [🔗]
[Ramaravind Kommiya Mothilal](#), Divyat Mahajan, Chenhao Tan, Amit Sharma
Proceedings of the 2021 AAAI/ACM Conference on AI, Ethics, and Society, 652-663 [AIES'21]
- [C.7] **Birds of a Caste-How Caste Hierarchies Manifest in Retweet Behavior of Indian Politicians** [🔗]
Palashi Vaghela*, [Ramaravind Kommiya Mothilal](#)*, Joyojeet Pal (* = Equal Contribution)
Proceedings of the ACM on Human-Computer Interaction [CSCW'21]

- [C.6] **Using Mobile Airtime Credits to Incentivize Learning, Sharing and Survey Response: Experiences from the Field** [🔗]
Devansh Mehta, [Ramaravind Kommiya Mothilal](#), Alok Sharma, William Thies, Amit Sharma
Proceedings of the 3rd ACM SIGCAS Conference on Computing and Sustainable Societies [COMPASS'20]
- [C.5] **Topical Focus of Political Campaigns and its Impact: Findings from Politicians' Hashtag Use during the 2019 Indian Elections** [🔗]
Anmol Panda, [Ramaravind Kommiya Mothilal](#), Monojit Choudhury, Kalika Bali, Joyojeet Pal
Proceedings of the ACM on Human-Computer Interaction [CSCW'20]
- [C.4] **Learnings from Technological Interventions in a Low Resource Language: A Case-Study on Gondi** [🔗]
Sebastin Santy, Devansh Mehta, [Ramaravind Kommiya Mothilal](#), Brij Mohan Lal Srivastava, Alok Sharma, Anurag Shukla, Vishnu Prasad, Venkanna U, Amit Sharma and Kalika Bali
International Conference on Language Resources and Evaluation [Invited Submission @ LRE Journal] [LREC'20]
- [C.3] **Explaining machine learning classifiers through diverse counterfactual explanations** [🔗]
[Ramaravind Kommiya Mothilal](#), Amit Sharma, Chenhao Tan
Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency [accompanying open-source library is part of Microsoft's Responsible AI Toolbox] [FAccT'20]
- [C.2] **Optimizing peer referrals for public awareness using contextual bandits** [🔗]
[Ramaravind Kommiya Mothilal](#), Amulya Yadav, Amit Sharma
Proceedings of the 2nd ACM SIGCAS Conference on Computing and Sustainable Societies [COMPASS'19]
- [C.1] **Predicting first-year engineering student success: from traditional statistics to machine learning** [🔗]
[Ramaravind Kommiya Mothilal](#), Tom Broos, Maarten Pinxten, Tinne De Laet
Proceedings of the 46th SEFI Annual Conference 2018 [SEFI'18]
- [W.3] **Evaluating Toxicity Understanding of LLM Agents** [🔗]
[Ramaravind Kommiya Mothilal](#), Syed Ishtiaque Ahmed, and Shion Guha
2025 IEEE International Conference on Collaborative Advances in Software and Computing [CASCON'25]
- [W.2] **Community-Driven Data Practices for Advancing Ethical and Equitable AI for Low-Resource Language** [🔗]
Charles Nimo, Shuheng Liu, Amy Z Chen, [Ramaravind Kommiya Mothilal](#), Michael Best
Proceedings of the ACM SIGCAS/SIGCHI Conference on Computing and Sustainable Societies [COMPASS'25]
- [W.1] **Towards Unifying Feature Attribution and Counterfactual Explanations**
[Ramaravind Kommiya Mothilal](#), Divyat Mahajan, Chenhao Tan, Amit Sharma
Workshop on Responsible AI, ICLR-21 [Spotlight Contributed Talk] [RAI@ICLR'21]
- [D.3] **Facilitating Media Distribution with Monetary Incentives** [🔗]
Devansh Mehta, Alok Sharma, [Ramaravind Kommiya Mothilal](#), Anurag Shukla, Vishnu Prasad, William Thies, Venkanna U, Colin Scott, Amit Sharma
Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems [CHI'20]
- [D.2] **Learnings from an Ongoing Deployment of an IVR-based Platform for Voter Awareness** [🔗]
[Ramaravind Kommiya Mothilal](#), Devansh Mehta, Alok Sharma, William Thies, Amit Sharma
Conference Companion Publication of the 2019 on Computer Supported Cooperative Work and Social Computing [a non-archival version won Outstanding Poster Award at COMPASS 2019] [CSCW'19]
- [D.1] **The Appeal of Influencers to the Social Media Outreach of Indian Politicians** [🔗]
Faisal M Lalani, [Ramaravind Kommiya Mothilal](#), Joyojeet Pal
Conference Companion Publication of the 2019 on Computer Supported Cooperative Work and Social Computing [COMPASS'19]

Leadership and Organizing

Founder & Organizer	Critical NLP Interest Group: Bringing interdisciplinary inquiry to NLP
Co-Organizer	FAccT'26 CRAFT workshop on Assumptions in Canada's Algorithmic Assessment
Co-Chair	COMPASS'25 Late Breaking Works
Mentor	2024 Responsible Tech Mentorship Program by All Tech Is Human
Co-Supervisor	UofT CS and EngSci undergraduate students: Joanna Roy and Jade Clement
Section Leader	<i>Code in Place</i> by Stanford University

Academic Service

Reviewer	Informal Logic Journal'26, CHI'26, CHI'25, FAccT'26, FAccT'25, CSCW'26, CSCW'25, CHI'24, DIS'24, SatML'24, ICLR-RAI'21, TOCHI'22, GHTC'22
Reviewer & PC	COMPASS'25, COMPASS'22, COMPASS'21, XKDD'21
Reviewer & AC	CHI'24 Late Breaking Works

Teaching

Teaching Assistant, UofT	Machine Learning with Applications in Python (INF 2179), Speculative Design (INF 2260), Experimental Design for Data Science (INF 2178), Computational Thinking (INF 1339), Programming for Data Science (INF 1340), Data Analytics (INF 2190), Data Modeling and Database Design (INF 1343), Human Values in Data Science (INF 2210), Computers and Society (CSC 300), Statistics for Data Science (INF 1344)
Workshop Presenter	Data Analysis with SQL and MySQL, iSkills Workshop at Faculty of Information, UofT (Fall 2022, Winter 2023, Fall 2023, Winter 2024)

Invited Talks

“Advancing Responsible AI in Practice” [slides]

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| › UofT AI Ethics Conference, Organized by IlluminAI, University of Toronto | Mar 2026 |
| › CS565: Ethics in AI, University of Illinois Urbana-Champaign | Apr 2025 |
| › Toronto Ethics in AI Symposium , University of Toronto | Jun 2024 |

“Caste Capital on Twitter: A Formal Network Analysis of Caste Relations” [slides]

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| › Department of Management Studies, Indian Institute of Technology Delhi | Jun 2022 |
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“When Machine Learning meets Educational Science” [slides]

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| › Faculty of Psychology and Educational Sciences, KU Leuven | Feb 2018 |
| › Leuven Engineering and Science Education Center (LESEC) | Mar 2018 |

Mentions and Coverages

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| › “Dalit MPs less likely to be heard on Twitter than upper castes” The Times of India | Oct 2020 |
| › “Birds of Brahmanical Feather Flock Together” The Wire | Aug 2020 |
| › “DiCE: Diverse Counterfactual Explanations for Hotel Cancellations” Towards Data Science | Jul 2020 |
| › “DiCE -ML models with counterfactual explanations for the sunk Titanic-” Analytics Vidhya | Mar 2020 |
| › “Open-source library provides explanation for machine learning through diverse counterfactuals” Microsoft Research Blog | Jan 2020 |
| › “Elections 2019: What tweet for tat tells you about Rahul Gandhi” LiveMint | Apr 2019 |